

A High-precision Current Detection Circuit for Battery Management System

PuSen Wu

Electronic Engineering College
Heilongjiang University
Harbin, China
1327486663@qq.com

Hao Xue

Electronic Engineering College
Heilongjiang University
Harbin, China
1051741664@qq.com

Byambajav Ragchaa

School of Integrated Circuits
Tsinghua University
Beijing, China
chengguang16@mails.tsinghua.edu.cn

Liji Wu*

School of Integrated Circuits
Tsinghua University
Beijing, China
lijiwu@tsinghua.edu.cn

Zhenhui Zhang*

Electronic Engineering College
Heilongjiang University
Harbin, China
Zhangzhenhui01@hlju.edu.cn

Xiangmin Zhang

School of Integrated Circuits
Tsinghua University
Beijing, China
zhxm@tsinghua.edu.cn

Abstract—This paper proposes a current detection circuit (CDC) for battery management systems(BMS), comprising a high-performance programmable gain amplifier (PGA) and a 16-bit high-precision, low-power Delta Sigma ADC. The PGA utilizes a two-stage folded cascode operational amplifier with resistive feedback to achieve adjustable gain. The ADC employs a single-loop second-order feedback modulator structure with an oversampling ratio of 256. This design enhances the conversion accuracy and reduces the output noise of the PGA and the switched-capacitor integrator using chopper stabilization techniques. The entire current detection circuit is implemented using the ASMC 0.35um BCD process. The post-simulation results demonstrate that under a 5V power supply voltage, the PGA maintains stable gain from 0-40dB with a phase margin greater than 60 degrees. The low-frequency noise of the PGA is measured at 1.386nV/sqrt(Hz). The transient simulation reveals that when the PGA input voltage varies within the range of 10mV to 300mV, it undergoes different levels of amplification to produce output voltages ranging from 0V to 5V. The PGA exhibits good power supply rejection ratio (PSRR) ranging from 94-112dB at 1KHz. For the ADC, with a signal bandwidth of 1KHz and a sampling rate of 1.024MHz, the effective number of bits (ENOB) reaches 16.58 bits, and the modulator operates at a current of 100uA. The simulation result shows that the error of the current detection is 0.3mV.

Keywords—BMS, PGA, Delta Sigma ADC, current detection

I. INTRODUCTION

With the rapid development of new energy vehicles, the battery management system (BMS) has become increasingly important[1]. To ensure battery safety, calculating the state of charge (SOC) of the battery is a crucial task for BMS. Based on the Coulomb sensing method, the SOC can be estimated by accurately monitoring the charging and discharging currents. The current amplitude can be derived by measuring the voltage generated across a shunt resistor using Ohm's Law. The current detection circuit (CDC) is illustrated in Figure 1. To achieve high accuracy in ADC full-scale conversion, a combination of programmable gain amplifier (PGA) and low-order incremental ADC is used to obtain accurate measurement results while minimizing heat generation[2].

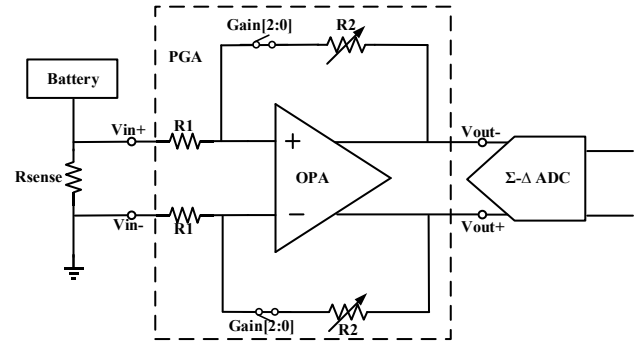


Fig. 1. Architecture of the CDC

The structure of this document unfolds in the subsequent manner: the exposition of current detection's system architecture is reserved for Section II, Section III will deliberate on the intricacies of circuit realization, encompassing both the PGA and the Delta Sigma ADC components. Section IV will be devoted to delineating the results of simulation, while the ultimate summation of this comprehensive paper shall reside within the confines of Section V.

II. SYSTEM ARCHITECTURE

A. PGA architecture

The overall structure of the PGA is shown in Figure 1. It utilizes a 3-to-8 decoder and transmission gate switches to control the programmable selection of the R2 resistor values through Gain[2:0]. The R2 resistor values are set at multiples of R1, specifically 1x, 2x, 4x, 8x, 16x, 32x, 64x, and 128x. The overall circuit's gain is determined by the negative feedback of the resistors as shown in equation (1).

$$AV = \frac{R_2}{R_1} \quad (1)$$

B. Delta sigma ADC architecture

Due to the limited bandwidth of the output voltage signal in the PGA, which is in the KHz range, a simple and stable single-loop second-order delta-sigma modulator is

employed to process the input voltage signal. The schematic diagram of the second-order feedback modulator used in this design is shown in Figure 2, consisting of two integrators, a one-bit quantizer, and a one-bit DAC circuit. With an oversampling ratio of 256, the structure employed in this design belongs to the Cascaded Integrator-Feedback (CIFB) structure, which has lower timing requirements. The input signal is simultaneously fed into the inputs of the first and second integrators, which helps reducing the output amplitude of the integrators, as demonstrated in the subsequent model simulations.

Herein, the parameter denoting the effect of the input signal takes the form of the signal transfer function (STF), elucidated in equation (2), while the factor governing the influence of noise is encapsulated within the noise transfer function (NTF), expounded upon in equation (3). The NTF serves to shift a significant portion of the quantization noise within the signal's frequency range towards higher frequencies, thereby dictating the overall stability of the modulator system. The distribution of zeros and poles in the NTF is visually depicted in Figure 3, and it is noteworthy that all poles reside comfortably within the confines of the unit circle, signifying the system's steadfast stability.

$$STF = \frac{L_0}{1-L_1} = \frac{B_2(Z-1) + B_1}{(Z-1)^2 + A_2(Z-1) + A_1} \quad (2)$$

$$NTF = \frac{1}{1-L_1} = \frac{(Z-1)^2}{(Z-1)^2 + A_2(Z-1) + A_1} \quad (3)$$

III. CIRCUIT IMPLEMENTATION

A. PGA schematic

The main operational amplifier circuit is shown in Figure 4. PMOS transistors are used as input pairs to obtain better frequency and noise performances. The main operational amplifier circuit consists of two stages. The first stage utilizes a folded cascode structure to increase the operational amplifier's gain. The second stage employs a common-source amplifier to amplify the output swing. Additionally, Miller compensation is used to compensate for zero and pole effects.

B. Chopper schematic

By introducing chopper stabilization before the input pair and cascode output as shown in Figure 5, the noise of the PGA can be reduced. The chopper stabilization technique is a frequency domain approach that modulates low-frequency noise and DC offset to higher frequency area[3], which can then be filtered out effectively. This process helps improve the overall performance of the PGA by mitigating the impact of low-frequency noise and offset, resulting in a cleaner output signal with reduced noise.

C. Modulator schematic

The Delta Sigma modulator plays a central role within the Delta Sigma ADC setup. It carries out the crucial tasks of oversampling and reshaping the analog input signal to minimize noise. In this particular design, a second-order feedback modulator is achieved through the use of a switched capacitor circuit. The comprehensive diagram, illustrated in Figure 6, encompasses the second-stage integrators, an adder, a one-bit quantizer, and a one-bit DAC circuit.

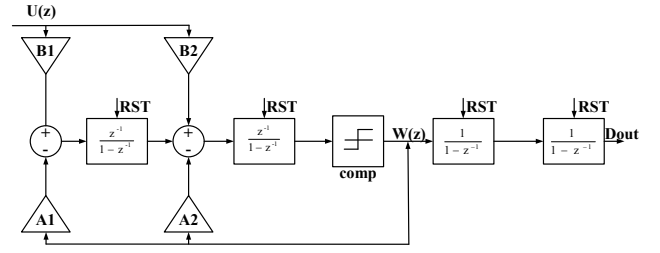


Fig. 2. Architecture of the modulator

D. Schematic of the integrator

The modulator circuit is implemented by switched-capacitor integrator which uses a fully-differential cascode chopper operational amplifier (PGA). The parameters of the integrator is determined by the ratio between capacitors. The input signal is passed through a cross-coupled switch before being fed to the inputs of the two integrators[4]. The control signals for the switches are generated using non-overlapping clock signals. The operational amplifier's offset can affect the conversion accuracy.

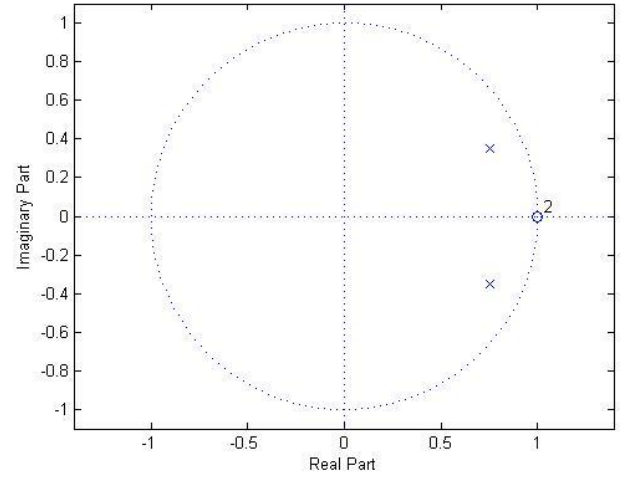


Fig. 3. Distribution of zeros and poles of NTF

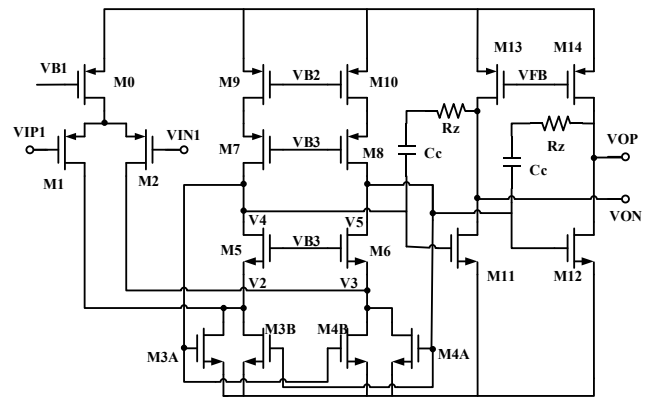
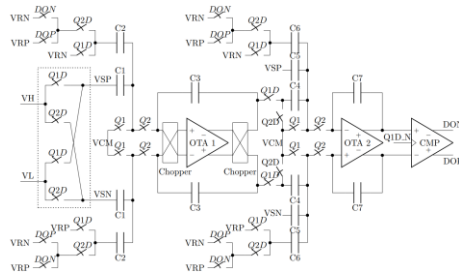
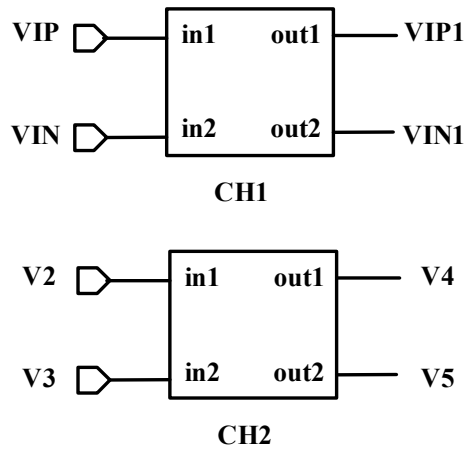
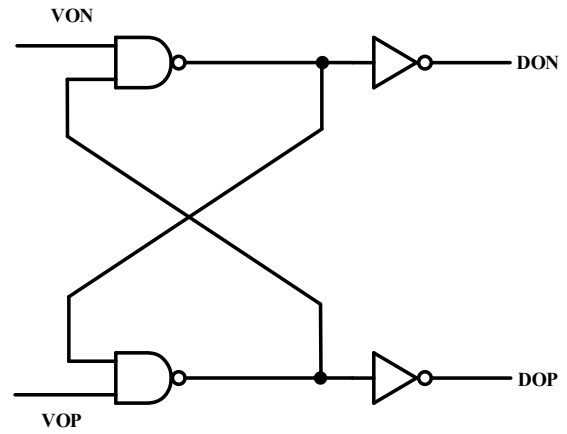
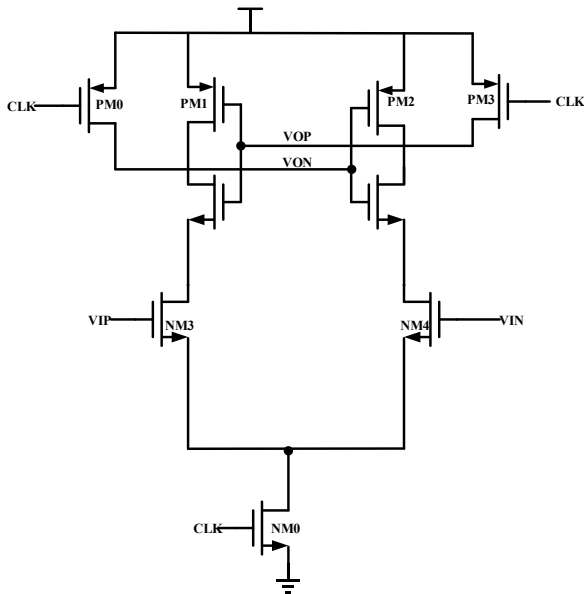


Fig. 4. Schematic of the PGA



E. Schematic of the one-bit quantizer

The one-bit quantizer in this design uses a dynamic comparator structure as shown in Figure 7 and Figure 8. When clk is high, the input signals VIP and VIN are compared, and the result of the comparison is stored using the SR latch .



IV. SIMULATION RESULTS

A. Simulation of the PGA

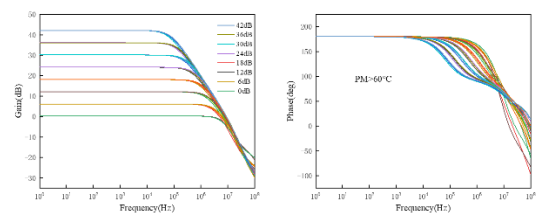
Figure 9 illustrates the frequency response of the programmable gain amplifier in the range of 0-40dB. The simulation results indicate that the bandwidth is greater than 100 kHz for all gain settings, and the phase margin is greater than 60° . These findings demonstrate that the amplifier exhibits good frequency characteristics across the entire gain range.

Figure 10 shows the power supply rejection ratio (PSRR) of the PGA at various gain settings. The simulation results show that the PSRR ranges from 97 dB to 123 dB at low frequencies, from 94 dB to 112 dB at 1 KHz, and from 66 dB to 114 dB at 1 MHz with eight gain settings at different corners.

Figure 11 indicates that, by adding sinusoidal signals of different amplitudes at the input of the PGA and amplifying them, the output can reach the full input range of the ADC, which is 0V to 5V with eight gain settings at different corners.

The output noise of the PGA under chopper stabilization is shown in Figure 12. It is evident that chopper stabilization helps suppress noise at low frequencies. In this paper, the PGA has a low output noise of $1.386\text{nV}/\sqrt{\text{Hz}}$, with eight gain settings at different corners.

In Figure 13, we can see the signal-to-noise ratio (SNR) obtained from time-domain post-simulation of the modulator. Different colors correspond to different temperatures and corners. The modulator was simulated over a period of 4096 points, corresponding to 7 cycles of the input waveform. The minimum SNR achieved is 99.8 dB, indicating a high level of precision exceeding 16 bits.



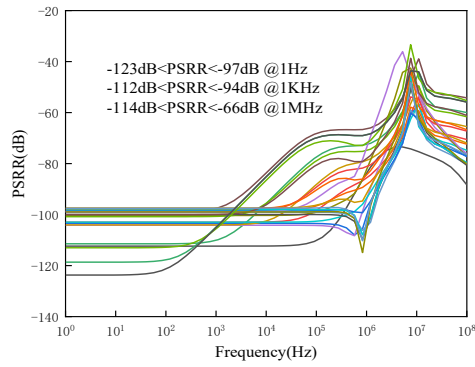


Fig. 10. Simulation of the PSRR plot

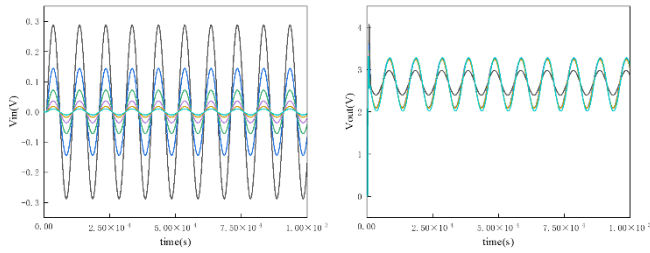


Fig. 11. Simulation of the VOUT/VIN plot

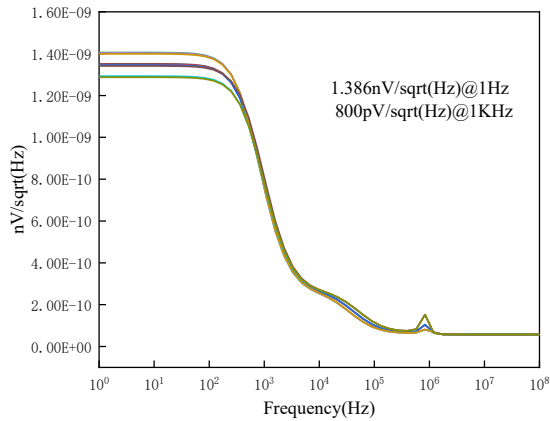


Fig. 12. Simulation of the NOISE plot

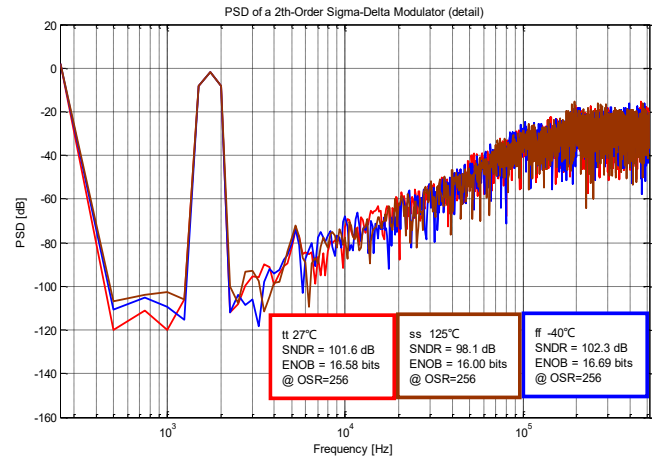


Fig. 13. Simulation of the SNR plot

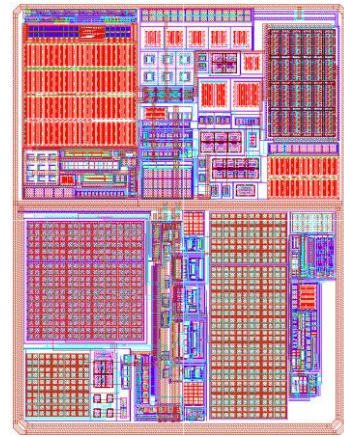


Fig. 14. Layout of the PGA and ADC

CONCLUSION

This paper presents the design of a current detection circuit using ASMC's 0.35um BCD process, intended for the front-end information collection chip in Battery Management Systems (BMS). The circuit is capable of handling a wide input voltage range, from 0.01V to 0.3V. The PGA utilizes a two-stage folded cascode operational amplifier with resistive feedback to achieve adjustable gain to get the output voltage, from 3V to 5V. The ADC employs a single-loop second-order feedback modulator structure with an oversampling ratio of 256. The layout is shown in Figure 14, and the modulator operates at a current of 100uA. The simulation result shows that the error of the current detection is 0.3mV.

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